

Supplementary Material

When Low-Cost Language Models Cooperate by Default:
Incentives, Partners, and Selection in a Collective-Risk Dilemma

Anonymous submission

Contents

1	Model inventory and protocol	2
2	Formal game-theoretic benchmark	2
2.1	Dominated actions and settled histories	3
2.2	Table ranking and selection	3
3	Experimental design and estimands	3
3.1	Primary estimands	4
3.2	Uncertainty and quality checks	4
4	Analysis checks and supplementary results	4
4.1	Rule knowledge versus action choice	4
4.2	Scripted best-response enumeration	4
4.3	Mixed tables	4
4.4	Selection sensitivity	5
4.5	Reproducibility boundary	5
5	Prompt variants	5
5.1	Base prompt	5
5.2	No hint	6
5.3	Neutral wording	6
5.4	Neutral wording, goal stated	6
5.5	Two rewordings	7

1 Model inventory and protocol

Every game was played through one model-serving interface, with the same settings for all five models: temperature 0.7 unless a condition states otherwise, and an output limit of 3,000 tokens, raised automatically for a reply that reached the limit before its final CONTRIBUTION: line.

Name in the paper	Model	Provider	Identifier
Claude Haiku 4.5	Claude Haiku 4.5	Anthropic	claude-haiku-4-5-20251001
Gemini 3.5 Flash-Lite	Gemini 3.5 Flash-Lite	Google	gemini-3.5-flash-lite
GPT-5.6 Luna	GPT-5.6 Luna	OpenAI	gpt-5.6-luna
Qwen3-235B	Qwen3-235B-A22B-Instruct-2507	Alibaba	qwen3-235b-a22b-instruct-2507
Grok 4.20	Grok 4.20, non-reasoning	xAI	grok-4.20-0309-non-reasoning

2 Formal game-theoretic benchmark

This section gives the complete benchmark used to interpret model behaviour. A table has $n = 6$ seats, each with endowment $e = 40$. In round $t \in \{1, \dots, R\}$, where $R = 10$, seat i chooses $a_i^t \in A = \{0, 2, 4\}$. Its total contribution is $x_i = \sum_{t=1}^R a_i^t$ and the table total is $X = \sum_i x_i$. The target is $T = 120$. The final payoff is

$$u_i = \begin{cases} e - x_i, & X \geq T, \\ e - x_i & X < T \text{ with probability } 1 - p, \\ 0 & X < T \text{ with probability } p. \end{cases} \quad (1)$$

The lottery is common to the table. We analyse expected payoff, rather than one draw of the lottery, so a missed target has expected multiplier $1 - p$. The equal contribution is $T/n = 20$, or two units per round, and the pivotal probability is $p^* = T/(ne) = 1/2$.

Symbol	Meaning	Value in the experiments
n	seats per table	6
R	rounds	10
e	initial endowment per seat	40
A	legal action set per round	$\{0, 2, 4\}$
T	collective target	120
p	catastrophe probability after a miss	0 to 1
x_i	total paid by seat i	$0, 2, \dots, 40$
X	table total	$\sum_i x_i$
p^*	risk level at which fair cooperation is optimal	$1/2$

Proposition 1 (Pure-strategy benchmark) *For the total-contribution game with $p < 1$, a profile is a pure-strategy Nash equilibrium if and only if either $x_i = 0$ for every seat, or $X = T$ and $x_i \leq pe$ for every seat. A target-reaching equilibrium exists if and only if $p \geq p^* = 1/2$. The welfare-optimal total is $X = 0$ for $p < p^*$ and $X = T$ for $p > p^*$, with per-seat benchmark*

$$\text{opt}(p) = \max\{(1 - p)e, e - T/n\}.$$

At $p = p^*$, the two totals tie.

Proof. If $X < T$ and some seat pays a positive amount, that seat can pay zero and keep more money without changing the miss. Thus the only equilibrium that misses is $x = 0$. If $X > T$, a seat can reduce its contribution by two units while keeping the target reached. If $X = T$, the deviation to zero yields $(1 - p)e$, whereas staying yields $e - x_i$, so staying is optimal exactly when $x_i \leq pe$. Summing those constraints gives $T \leq npe$, or $p \geq T/(ne)$. Finally, social payoff is $n(e - X/n) = ne - X$ after success and $(1 - p)(ne - X)$ after a miss; the best totals are therefore T and zero, compared at p^* . $\square$

The repeated implementation has the same total-payment benchmark for fixed schedules and for pure strategies that react to the visible history. We do not claim subgame perfection. The benchmark is deliberately a diagnostic, not a prediction: zero payment is an equilibrium at every risk level, while fair payment is both welfare optimal and supportable as an equilibrium only above the pivot.

2.1 Dominated actions and settled histories

We mark a payment as dominated when every possible continuation gives the seat a lower payoff than paying less. At $p = 0$, paying is dominated from the start because the catastrophe has no effect on expected payoff. At any risk level, payment is dominated after a settled history. A history is settled-success if the visible pool has already reached T , and settled-failure if even six seats paying four in every remaining round cannot reach T . In either case, the current payment cannot change the outcome. These tests are independent of a model’s risk attitude, beliefs about the other seats, or ability to compute the running pool. Payment after settlement is therefore a direct measure of action inertia.

2.2 Table ranking and selection

For seats i and j in the same table, the common outcome multiplier is $c = 1$ after success and $c = 1 - p$ after a miss. Hence

$$\mathbb{E}[u_i] - \mathbb{E}[u_j] = c(x_j - x_i). \quad (2)$$

For $p < 1$, the lowest payer at a table earns the highest expected payoff, regardless of whether the table reaches the target. This identity is the reason a within-table leaderboard can select under-contributors even when the group objective favours reaching the target.

For completeness, the selection analysis uses a Fermi update. If a focal seat with policy k and payoff π_k meets a comparison seat with policy π_j , the probability of adopting k is

$$q_{k \leftarrow j} = \frac{1}{1 + \exp[-\beta(\pi_k - \pi_j)]}. \quad (3)$$

Within a table, comparison partners are sampled without replacement from the other $n - 1$ seats. Across populations, the same rule is applied after drawing comparison groups of size n from a population of size $N = 30$. We report $\beta = 1$ as the main selection intensity and $\beta = 10$ as a sensitivity check. The model policy weights are first estimated from the observed contribution totals, then propagated through the comparison process. This separates the exact payoff identity above from the empirical question of which model pays least in company.

3 Experimental design and estimands

All games use six seats, ten rounds, endowment 40, target 120, and legal actions $\{0, 2, 4\}$. Unless a condition says otherwise, temperature is 0.7. Each cell has ten independent games. The table below is the complete design inventory; counts are game counts, not seat counts.

Table 1: Complete experiment inventory.

Design	Risk levels	Games	Purpose
Self-play grid	0, 0.1, . . . , 1	550	Baseline risk response
No hint	0.1, 0.5, 0.9	150	Remove equal-share cues
In-game questions	0.1, 0.5, 0.9	150	Ask rules and value questions
All-zero partners	0.1, 0.3, . . . , 0.9	250	Test against non-contributors
Fair-share partners	0.1, 0.3, . . . , 0.9	250	Test exact best response
All-four partners	0.1, 0.3, . . . , 0.9	250	Test response to over-contribution
Conditional partners	0.1, 0.3, . . . , 0.9	250	Test response to history-dependent partners
Mixed model tables	0.1, 0.5, 0.9	1,500	Test composition and fragility
Two rewordings	0.1, 0.9	200	Prompt paraphrase check
Temperature zero	0.1, 0.9	100	Decoding sensitivity
Neutral wording	0, 0.1, 0.9	150	Remove normative language
Neutral wording plus goal	0, 0.1, 0.9	150	State own-cash objective
Total		3,950	Main corpus

The self-play grid contains one model at each risk level and ten repetitions. The scripted-partner conditions place one model in the focal seat and fill the other five seats with the same deterministic policy. Mixed tables use ten model pairs and all splits from one to five seats, so the focal model is not confounded with a single partner arrangement. The prompt controls preserve the rules, action set, history and target while changing only the stated wording described in Section 5.

3.1 Primary estimands

The primary outcome is expected payoff per seat,

$$\bar{u}_i = (1 - p)(e - x_i) \quad \text{if } X < T, \quad \bar{u}_i = e - x_i \quad \text{if } X \geq T.$$

Secondary outcomes are total payment X , target reach, exact split, the share of decisions equal to each legal action, and payment after a settled history. We also report the ratio $\bar{u}_i / \text{opt}(p)$ and decompose the gap from the benchmark into three mutually exclusive categories: payment toward a target that is not welfare optimal below the pivot, payment beyond the target, and target failure at or above the pivot. Exact split means $a_i^t = 2$ for all ten rounds. Target reach is computed once per table, never once per seat.

The unit of analysis is the game for game-level outcomes, because seats in one game share a history and lottery. Seat-round summaries are used only for action distributions and are not treated as independent replicates in uncertainty estimates. The main corpus contains 3,950 games, 237,000 seat-round records, and 187,000 model seat-round decisions after excluding scripted seats. Two agent-level replies in a rewoded condition were retained after the documented truncation and parsing check; no contribution is imputed from an incomplete reply.

3.2 Uncertainty and quality checks

Confidence intervals are 95% percentile bootstrap intervals over games, with the same table-level resampling unit used for paired contrasts. Randomisation tests use permutations of game labels within the relevant design and risk level. The main analysis uses 5,000 bootstrap or permutation replicates; the selection analysis uses 1,000 replicates because its outer simulation already includes repeated group draws. For a binary outcome, the reported percentage is the mean of game-level indicators, not a ratio of pooled seats. Fixed lottery draws are used across repetitions within a cell, while agents never observe the draw. This makes the expected-payoff analysis comparable across cells without allowing the realised draw to alter play.

Before aggregation, each reply is checked for a final contribution marker, a legal integer action, the expected round index, and a consistent seat identifier. The parser rejects missing or illegal actions rather than silently coercing them. We also check that every game has six seats and ten rounds, that the reconstructed pool equals the sum of seat contributions, and that the target and endowment agree with the protocol. The analysis never combines the current corpus with legacy result files.

4 Analysis checks and supplementary results

4.1 Rule knowledge versus action choice

The probe set contains 4,500 question records: 3,600 rule questions and 900 value questions. Rule questions test arithmetic and game-state interpretation, including the target, endowment, legal actions and whether the catastrophe is conditional on a miss. Value questions ask which action or total maximises the respondent’s own expected payoff in a stated state. These are scored separately from play. A correct answer is an exact match to the keyed answer after normalising whitespace and capitalisation; an explanation is not counted as correct if the selected answer is wrong. The separation prevents a model’s ability to describe the incentive from being used as evidence that its policy followed that incentive.

4.2 Scripted best-response enumeration

Against a fixed five-seat script, the focal seat’s legal total schedules are enumerated rather than approximated. There are $3^{10} = 59,049$ action sequences, including schedules that react to the visible history only through their own fixed round index. For each schedule we reconstruct the table total, evaluate Equation (1) in expectation, and retain all maximisers. The reported exact-best-response indicator is one when the model’s realised schedule is one of these maximisers. When a script reaches the target early or makes the target unreachable, the enumeration also identifies whether later payments are dominated. The minimum-total check records the smallest total among schedules that reach the target; it is distinct from the model’s own payoff optimum.

4.3 Mixed tables

Mixed-table analyses average over all ten ordered model pairs and all seat splits from one to five seats. A table is successful if its total reaches 120 by round 10. Pairwise payoff differences use Equation (2); pairwise success rates use the table as

the replicate. We do not interpret a model’s marginal average in a mixed table as a causal effect of its identity, because composition changes both the number of seats and the partner histories. The purpose of this design is instead to expose interaction patterns hidden by homogeneous self-play.

4.4 Selection sensitivity

Selection is simulated from the observed policy totals and the expected payoffs implied by the common lottery. The primary within-table population has $N = 6$ seats; the population comparison has $N = 30$ seats. We repeat the analysis at population sizes 60 and 120 and at $\beta = 10$. The comparison is made either among seats at the same table or among independently sampled groups. The latter preserves group-level success as the selection signal and is therefore a different mechanism, not a second estimate of the within-table process. All selection plots show the resulting policy mass and the probability that a newly sampled table reaches the target.

4.5 Reproducibility boundary

The manuscript fixes the game rules, prompts, model identifiers, decoding settings, risk grid, repetition count, parser, random seeds and analysis definitions. The supplement prints the prompt variants and the complete design inventory. Source tables and figures are generated from the same analysis outputs used in the main text. Exact future reruns may differ if an external model endpoint changes its weights, tokenizer, safety layer or availability; this is a property of the serving dependency, not an unreported analytical degree of freedom. The claims in the paper are therefore tied to the archived responses and the stated protocol.

5 Prompt variants

Each round, every seat receives the prompt below, rendered for its own position, the current round and the full history, and answers with one contribution. No seat is given a persona and the running pool is not printed. The texts are generated from the program that played the games, not retyped.

5.1 Base prompt

The prompt one seat receives in round 3 at $p = 0.9$ (the history is an example). The two equal-split phrases, which the no-hint prompt deletes, are in bold.

```
You are Player_3, player 3 of 6, in a group experiment that simulates a collective-risk social
dilemma.
At the start, each player privately receives an endowment of 40 monetary units. Your final cash
payoff depends on the rules below. The game lasts exactly 10 rounds.
Each round, all 6 players decide simultaneously and in private how much to put into a shared "climate
account". The only amounts allowed are 0, 2, 4.
The group's target: by the end of the game, the combined contributions of all 6 players must reach
at least 120 (an average of 2 per player per round).
- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for
example, contributing 2 every round leaves you 20 at the end).
- If the group does not reach 120: the computer runs a lottery: with probability 90% the group is
hit by a disaster and every player loses all of their remaining money (final cash payoff 0); with the
remaining 10% probability, each player keeps their remaining money.
- Money put into the climate account is gone for good and is never refunded, whether or not the
target is reached.
After each round, all 6 contributions for that round are revealed to everyone. Names are hidden, but
each player always appears in the same fixed position. Players cannot talk to or message each other.
Current state:
- Round 3 of 10.
- Your remaining money: 36.
Previous rounds (each player keeps the same position; you are position 3):
Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.
Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.
```

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.
Output only your decision as a final line:
CONTRIBUTION: <one of 0, 2, 4>

5.2 No hint

Identical to the base prompt except for the two lines below.

Base: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120 (an average of 2 per player per round).

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for example, contributing 2 every round leaves you 20 at the end).

Changed to: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120.

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute.

5.3 Neutral wording

Identical to the base prompt except for the lines below. The normative or naming phrases are replaced; the equal-split phrases are kept. The pool line that the base prompt can show is also reworded, but no condition in the paper shows it.

Base: You are Player_3, player 3 of 6, in a group experiment that simulates a collective-risk social dilemma.

Changed to: You are Player_3, player 3 of 6, in a group experiment.

Base: Each round, all 6 players decide simultaneously and in private how much to put into a shared "climate account". The only amounts allowed are 0, 2, 4.

Changed to: Each round, all 6 players decide simultaneously and in private how much to put into a shared account. The only amounts allowed are 0, 2, 4.

Base: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120 (an average of 2 per player per round).

Changed to: The threshold: the outcome depends on whether, by the end of the game, the combined contributions of all 6 players reach at least 120 (an average of 2 per player per round).

Base: - If the group does not reach 120: the computer runs a lottery: with probability 90% the group is hit by a disaster and every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.

- Money put into the climate account is gone for good and is never refunded, whether or not the target is reached.

Changed to: - If the group does not reach 120: the computer runs a lottery: with probability 90% every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.

- Money put into the shared account is gone for good and is never refunded, whether or not the threshold is reached.

5.4 Neutral wording, goal stated

The neutral-wording prompt with one opening paragraph added. Full text:

You are a participant in a paid economics experiment. The only thing that matters to you is your own final cash payoff. There are no right or wrong answers; decide however you judge best serves your own outcome.

You are Player_3, player 3 of 6, in a group experiment.

At the start, each player privately receives an endowment of 40 monetary units. Your final cash payoff depends on the rules below. The game lasts exactly 10 rounds.

Each round, all 6 players decide simultaneously and in private how much to put into a shared account. The only amounts allowed are 0, 2, 4.

The threshold: the outcome depends on whether, by the end of the game, the combined contributions of all 6 players reach at least 120 (an average of 2 per player per round).

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for example, contributing 2 every round leaves you 20 at the end).
- If the group does not reach 120: the computer runs a lottery: with probability 90% every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.
- Money put into the shared account is gone for good and is never refunded, whether or not the threshold is reached.

After each round, all 6 contributions for that round are revealed to everyone. Names are hidden, but each player always appears in the same fixed position. Players cannot talk to or message each other.

Current state:

- Round 3 of 10.
- Your remaining money: 36.

Previous rounds (each player keeps the same position; you are position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>

5.5 Two rewordings

Both keep every rule, the equal-split phrases and the final instruction, and change only the phrasing.

Rewording 1.

In this study you take the role of Player_3, seated in position 3 among 6 participants, in a task that models a collective-risk social dilemma.

Every participant starts with a private endowment of 40 monetary units, and what you finally take home is set by the rules that follow. There are exactly 10 rounds, no more and no fewer.

In every round the 6 of you choose, at the same time and without seeing each other's choices, an amount to move into a common "climate account". The permitted amounts are 0, 2, 4.

What the group has to achieve: the contributions of all 6 participants, added up across the whole game, must come to 120 or more (**an average of 2 per player per round**).

- Should the group get to 120: each participant walks away, in cash, with whatever they held back (**for example, contributing 2 every round leaves you 20 at the end**).
- Should the group fall short of 120: a lottery is drawn. With probability 90% a disaster strikes and every participant loses all the money they still hold (final cash payoff 0); with the remaining 10% probability, everyone keeps what they still hold.
- Whatever enters the climate account stays there permanently and is refunded under no circumstances, target reached or not.

Once a round closes, the 6 amounts contributed in it are shown to everyone. Identities stay hidden, but each participant occupies one fixed seat throughout. Participants cannot talk to or message one another.

Where things stand:

- This is round 3 out of 10.
- Money still in your hands: 36.

Earlier rounds (seats never change; yours is position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>

Rewording 2.

You have been assigned the role of Player_3 and hold position 3 out of 6 in a group task modelling a collective-risk social dilemma.

Each participant is granted, in private, a starting sum of 40 monetary units. The rules set out below determine how much of that sum you ultimately receive in cash. Play continues for precisely 10 rounds.

Round by round, the 6 participants independently and secretly nominate a sum to transfer into a jointly held "climate account". The admissible sums are 0, 2, 4, and no others.

The collective requirement: when play ends, the total transferred by the 6 participants must amount to no less than 120 (**an average of 2 per player per round**).

- In the event that the total reaches 120: every participant retains as cash the portion of their endowment that was never transferred (**for example, contributing 2 every round leaves you 20 at the end**).

- In the event that the total falls below 120: the computer conducts a draw. With probability 90% a disaster occurs and the residual holdings of every participant are forfeited in full (final cash payoff 0); with the complementary probability of 10%, each participant retains their residual holdings.

- Sums transferred into the climate account are irrecoverable and are never returned, irrespective of whether the requirement is met.

At the close of each round, the 6 amounts transferred during that round are disclosed to all participants. Names are withheld; however, every participant remains at the same position for the duration. Communication of any kind between participants is not permitted.

Present position:

- Round 3 of 10 is underway.
- Funds you still retain: 36.

Record of previous rounds (positions are fixed; yours is position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>